Statistical Physicsz

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Índice general

Capítulo 1

Postulates of Statistical Physics. Ensembles

We have two ways to measure a property A (discussed previously). For example, the temperature of a bottle of water. The first way is measure the same bottle several times in different moments, in every t_N we measure A_N, then property A will be

$$\bar{A} = \lim_{t_N \to \infty} \int_0^{t_N} A(\vec{q}, \vec{p}) dt \tag{1.1}$$

the other option is measure N samples, N bottles of water. The measured value for each samples is A_N, then the property A will be

$$\langle A \rangle = \sum_{i=0}^{N} P_i A_i(\vec{q}, \vec{p}) \tag{1.2}$$

Nota

Both are equivalent if our system is ergodic, we will see more details later

where P_i is the probability to measure the value A_i, generally $P_i = 1/N$ for equal energies. The integral definition is

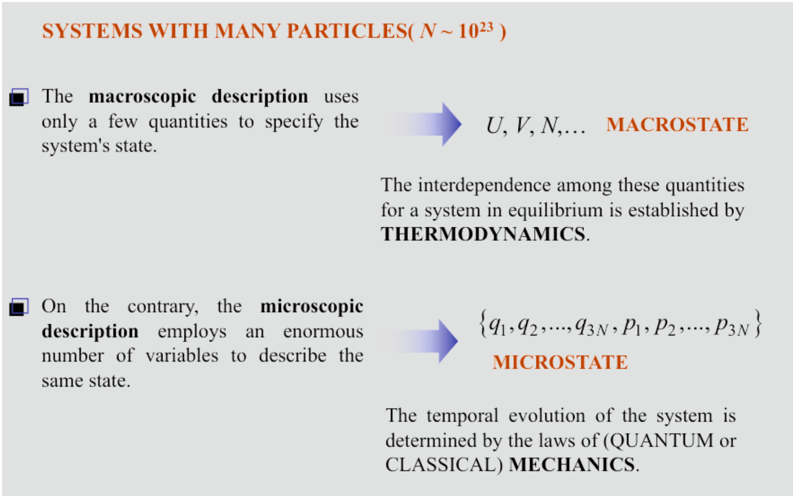
$$\langle A \rangle = \int_\Gamma \rho(\vec{q}, \vec{p}) A(\vec{q}, \vec{p}) d\vec{q} d\vec{p} \tag{1.3}$$

with ρ normalised

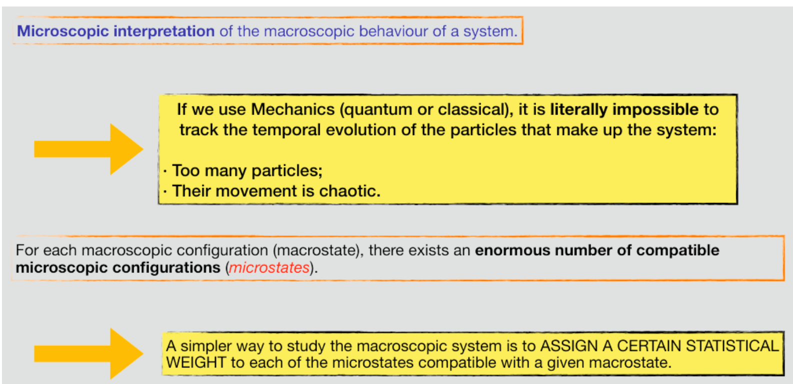
$$\int_\Gamma \rho(\vec{q}, \vec{p}) d\vec{q} d\vec{p} = 1 \tag{1.4}$$

1.1. Definition of Ensemble

1.1.1. Systems with Many Particles ($N \sim 10^{23}$)



In the study of systems with an enormous number of particles, typically on the order of 10^{23} or more, we adopt a distinct approach to understand their behaviour. This approach allows us to focus on the **macroscopic aspects** of these systems, where only a few key variables are needed to define the state of the system. These essential macroscopic variables, such as the internal energy U, volume V, and the number of particles N, encapsulate the system's conditions. How these variables are interconnected for a system in equilibrium is meticulously described by the principles of **thermodynamics**. In this ensemble view, we consider an ensemble of microstates, each characterised by a specific set of positions and momenta for all particles. The collection of these **microstates** represents a microcanonical ensemble, encompassing all conceivable microstates that correspond to the given macrostate. These microstates are determined by the positions $\{q_1, q_2, \ldots, q_{3N}\}$ and momenta $\{p_1, p_2, \ldots, p_{3N}\}$ of all particles, totalling $3N$ variables.

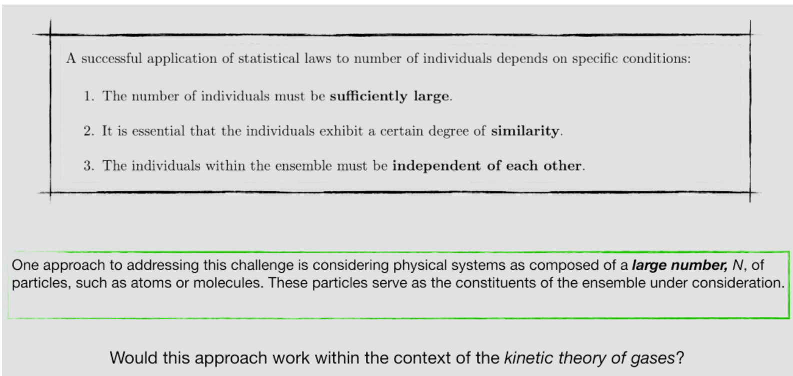


The system's evolution over time follows the dynamical laws of either **quantum mechanics** or **classical mechanics**, depending on the scale and nature of the system. While thermodynamics guides us in understanding the macroscopic equilibrium properties, the microscopic details are shaped by the principles of mechanics, whether they are classical or quantum in nature. This interplay between macroscopic and microscopic descriptions forms the foundation of statistical physics for systems with a vast number of particles. Microscopically interpreting the macroscopic behaviour of a system involves several challenges when using Mechanics, whether in the quantum or classical context. Tracking the temporal evolution of particles within the system becomes virtually **impossible** due to two key factors:

- **Enormous particle count**: the system typically consists of a vast number of particles, making individual tracking impractical;
- **Chaotic motion**: the motion of these particles is often chaotic, further complicating any attempts at precise tracking.

For each macroscopic configuration or macrostate, there exists a vast number of compatible microscopic configurations or microstates. This multitude of microstates is a fundamental aspect of the microscopic description of the system. To facilitate the study of the macroscopic system's behaviour, a more straightforward approach is adopted. This approach assigns a particular **statistical weight** to each of the microstates that are consistent with a given macrostate.

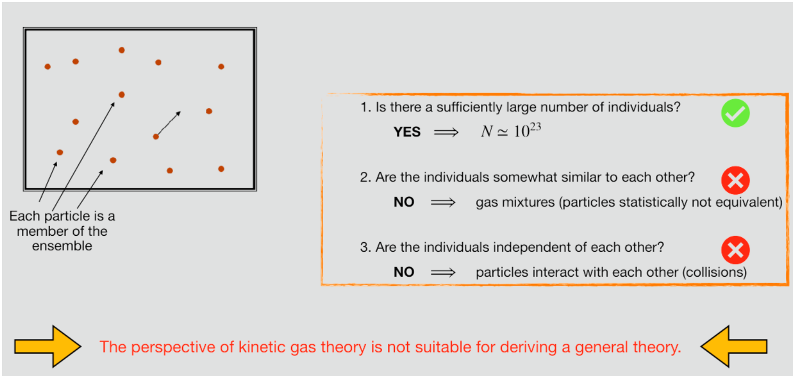
1.1.2. Satisfactory Application of Statistical Laws to an Ensemble of Individuals



A successful application of statistical laws to number of individuals depends on specific conditions:

1. The number of individuals must be **sufficiently large**.
2. It is essential that the individuals exhibit a certain degree of **similarity**.
3. The individuals within the ensemble must be **independent of each other**.

One approach to addressing this challenge is considering physical systems as composed of a large number, N, of particles, such as atoms or molecules. These particles serve as the constituents of the ensemble under consideration. Can one use the **kinetic theory of gases** to this end?



The assumption underlying this approach is that, when dealing with a substantial number of particles, the collective behaviour of these particles, while still following the principles of statistical physics, can be analysed effectively without explicitly tracking each individual particle. The application of statistical laws to such systems, when the conditions mentioned above are met, provides valuable insights into the macroscopic properties and behaviour of these systems. In the context of the kinetic theory of gases, each particle is considered a constituent of the ensemble. Whether this application of statistical physics is satisfactory depends on the following criteria:

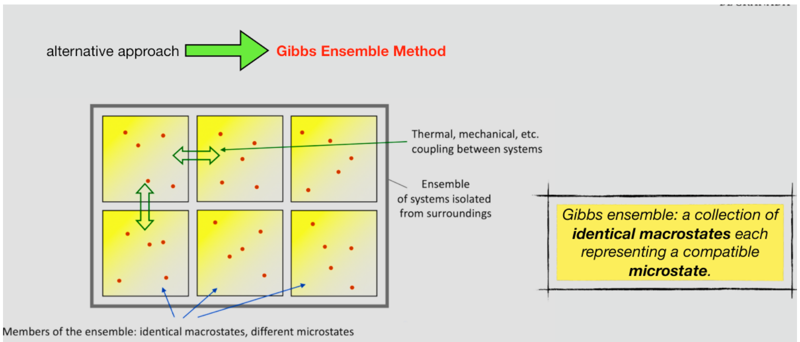
1. Is there a sufficiently large number of individuals?
2. Are the individuals somewhat similar to each other?
3. Are the individuals independent of each other?

When addressing these questions in the context of the kinetic theory of gases, we find the following:

1. Indeed, the number of particles in a macroscopic system is extremely large, typically on the order of $N \sim 10^{23}$.
2. The similarity of particles is a requirement for successful statistical analysis. However, in the case of **gases composed of more than one component**, the particles are no longer equivalent from a statistical perspective.
3. The independence of particles is a fundamental consideration. Unless dealing with a diluted gas, **interactions** among particles are common, affecting gases, liquids, solids, and more.

Therefore, in the context of statistical physics, the kinetic theory of gases falls short in providing a comprehensive general theory. An alternative approach, known as the **Gibbs Ensemble Method**, introduced by Gibbs in 1902, offers a more suitable perspective for the development of a broader theoretical framework. This method transcends the limitations of the kinetic theory of gases and holds significant promise for addressing complex systems in the realm of statistical physics.

1.1.3. Gibbs Ensemble Method: What Is It?



A **Gibbs Ensemble** is a concept that involves the coupling of multiple systems, each of which can exhibit various behaviours. These systems can be thermally, mechanically, or materially linked in various ways. When considering a Gibbs Ensemble, the focus is on an isolated collection of systems that share **identical macrostates** but possess different microstates. In essence, within a Gibbs Ensemble, multiple macroscopically identical systems are analysed, each characterised by its **microstates**. The **probability** (P_n) of a system residing in a specific microstate n, which is consistent with the macrostate, is a key parameter in this method. These probabilities provide valuable insights into the behaviour of systems within the ensemble.

Now, with reference to figure of above let's assess whether the three conditions are met:

1. **Are there a sufficiently large number of individuals?** The number of systems in the ensemble can be made arbitrarily large: $N_c \to \infty$.

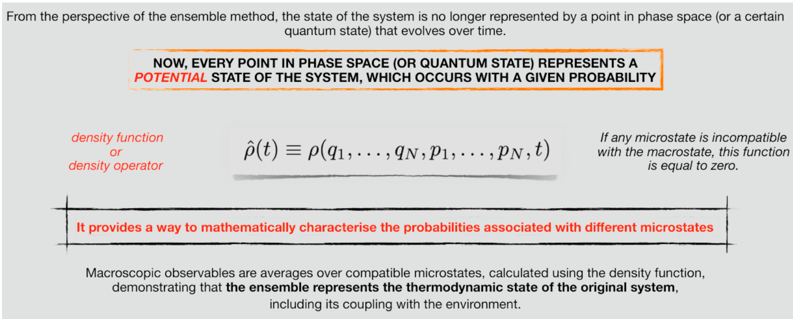
2. **Are the individuals *similar* to each other?** Yes, they are. The systems in the ensemble are macroscopically identical.

3. **Are the individuals *independent* of each other?** If a system interacts with others solely through its surface, the interaction energy is negligible compared to the system's energy. Therefore, each system is independent:

$$E = E_1 + E_2 + E_{12} \approx E_1 + E_2 \tag{1.5}$$

1.1.4. First Postulate: Postulate of Averages

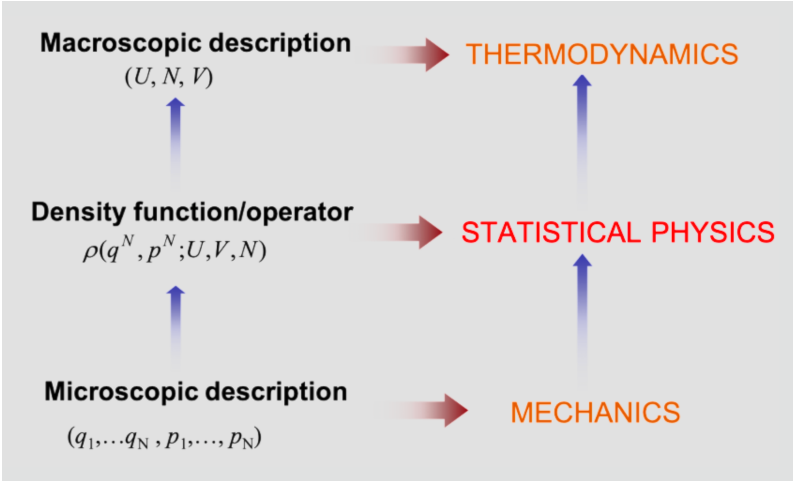
Density function/operator



In the context of the ensemble method, the conventional portrayal of a system's state as a point in phase space (or a specific quantum state) evolving over time undergoes a substantial change of perspective. In this framework, each point within the phase space (or quantum state) now serves as a representation of a **potential** state of the system, which occurs with a given probability. This change in perspective introduces the concept of a **density function** or **density operator**. This operator, pivotal in the statistical description of a system, reads

$$\hat{\rho}(t) \equiv \rho(q_1, \ldots, q_N, p_1, \ldots, p_N, t) \tag{1.6}$$

It provides a way to mathematically characterise the probabilities associated with different microstates. If any microstate is incompatible with the macrostate, this function is equal to zero. The idea of a density operator is central to this framework. The density operator is a mathematical construct used to connect the macroscopic observables with the myriad of microstates a system can access. Through the density operator, we can calculate averages of various physical properties, enabling us to describe the system's behaviour in statistical terms. In other words, the macroscopic observables are obtained as averages over all compatible microstates, calculated using the density function (operator). This approach relies on the assumption that the systems forming the collectivity of microstates collectively represent the thermodynamic state and the interactions with the external environment of the original system. This ensures that the statistical properties of the microstates adequately reflect the macroscopic observables. In summary, the transition from macroscopic thermodynamics to the microscopic world of statistical physics is facilitated by the density function ρ, connecting the statistical behaviour of a system with its fundamental microstate descriptions).



In the realm of thermodynamics, a macroscopic description of a system is characterised by internal energy (U), number of particles (N), and volume (V). These thermodynamic variables provide a framework for understanding and analysing the behaviour of macroscopic systems. To delve into the statistical aspects of physical systems, we shift our perspective to consider the density function (or operator) denoted by $\rho(q^N, p^N; U, V, N)$. This function connects the macroscopic parameters (U, N, V) with the microscopic descriptions, allowing us to explore the statistical behaviour of the system. At the level of microscopic descriptions, we analyse the system in terms of individual positions and momenta: $(q_1, \ldots, q_N, p_1, \ldots, p_N)$. This microscopic perspective aligns with the principles of mechanics, enabling us to scrutinise the detailed dynamics and interactions at the particle level.

1.2. Classical Formulation of the First Postulate

Macroscopic Physical Quantities. These are represented by functions of space and time coordinates, often taking the form of fields: $A(\mathbf{r}, t)$. Thermodynamics, in particular, focuses on equilibrium states: $A(\mathbf{r}, t) \equiv A(\mathbf{r})$.

Microscopic Physical Quantities. In contrast, these are described by dynamic functions in phase space coordinates $\{\mathbf{q}, \mathbf{p}\} \equiv \{q_1, \ldots, q_N, p_1, \ldots, p_N\}$. Macroscopic quantities can be simplified as functions of their spatial coordinates only: $a(\mathbf{q}, \mathbf{p}; \mathbf{r})$

1.2.1. Classical Formulation of the First Postulate

First Postulate

"The values of macroscopic physical quantities at time t are given as the averages of microscopic dynamic functions."

$$A(\mathbf{r}, t) \equiv \langle a(\mathbf{q}, \mathbf{p}; \mathbf{r}) \rangle = \int_\Gamma a(\mathbf{q}, \mathbf{p}; \mathbf{r}) \rho(\mathbf{q}, \mathbf{p}; t) \, d\mathbf{q} \, d\mathbf{p}$$

$$\int_\Gamma \rho(\mathbf{q}, \mathbf{p}; t) \, d\mathbf{q} \, d\mathbf{p} = 1 \quad \text{Normalization} \quad \rho(\mathbf{q}, \mathbf{p}; t) \geq 0$$

The First Postulate (Classical Formulation): "The values of macroscopic physical quantities at time t are given as the averages of microscopic dynamic functions."

$$A(\mathbf{r}, t) \equiv \langle a(\mathbf{q}, \mathbf{p}; \mathbf{r}) \rangle = \int_{\Gamma} a(\mathbf{q}, \mathbf{p}; \mathbf{r}) \rho(\mathbf{q}, \mathbf{p}; t)\, d\mathbf{q}\, d\mathbf{p} \tag{1.7}$$

$$\int_{\Gamma} \rho(\mathbf{q}, \mathbf{p}; t)\, d\mathbf{q}\, d\mathbf{p} = 1 \quad \text{Normalization} \quad \rho(\mathbf{q}, \mathbf{p}; t) \geq 0 \tag{1.8}$$

Here, $\rho(\mathbf{q}, \mathbf{p}; t)$ represents the density function and $\rho(\mathbf{q}, \mathbf{p}; t)\, d\mathbf{q}\, d\mathbf{p}$ is the probability of finding a microstate within the hypervolume $d\mathbf{q}\, d\mathbf{p}$ located at the point $(\mathbf{q}, \mathbf{p})$ at time t. A good example is offered by the relation between energy and Hamiltonian of a system:

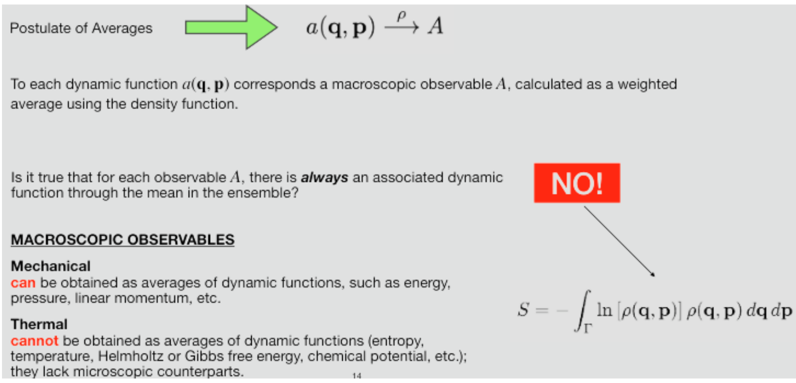
$$E = \langle H(\mathbf{q}, \mathbf{p}) \rangle = \int_{\Gamma} H(\mathbf{q}, \mathbf{p}) \rho(\mathbf{q}, \mathbf{p})\, d\mathbf{q}\, d\mathbf{p}$$

Here, $\rho(\mathbf{q}, \mathbf{p}; t)$ represents the density function and $\rho(\mathbf{q}, \mathbf{p}; t)\, d\mathbf{q}\, d\mathbf{p}$ is the probability of finding a microstate within the hypervolume $d\mathbf{q}\, d\mathbf{p}$ located at the point $(\mathbf{q}, \mathbf{p})$ at time t. A good example is offered by the relation between energy and Hamiltonian of a system:

$$E = \langle H(\mathbf{q}, \mathbf{p}) \rangle = \int_{\Gamma} H(\mathbf{q}, \mathbf{p}) \rho(\mathbf{q}, \mathbf{p})\, d\mathbf{q}\, d\mathbf{p} \tag{1.9}$$

where E represents the average energy of a system; $\langle H(\mathbf{q}, \mathbf{p}) \rangle$ denotes the average value of the system's Hamiltonian, which describes the total energy of the system; $H(\mathbf{q}, \mathbf{p})$ is the Hamiltonian function, which depends on the positions $\mathbf{q}$ and momenta $\mathbf{p}$ of the system's microscopic constituents (*e.g.*, particles) and captures the energy of the system in terms of these microscopic variables; and $\rho(\mathbf{q}, \mathbf{p})$ is the probability density function for the system, which characterises the likelihood of finding the system in a specific microstate defined by positions $\mathbf{q}$ and momenta $\mathbf{p}$. This equation expresses that the average energy E of the system is determined by calculating the expected value of the Hamiltonian using the probability density function $\rho(\mathbf{q}, \mathbf{p})$. In other words, it relates the macroscopic property (average energy) to the microscopic properties (positions and momenta of the particles) and their associated probabilities.

1.2.2. Mechanical and Thermal Magnitudes



Postulate of Averages. Every dynamic function $a(\mathbf{q},\mathbf{p})$ corresponds to a macroscopic observable A, calculated as an average weighted by the density function.

$$a(\mathbf{q},\mathbf{p}) \xrightarrow{\rho} A \tag{1.10}$$

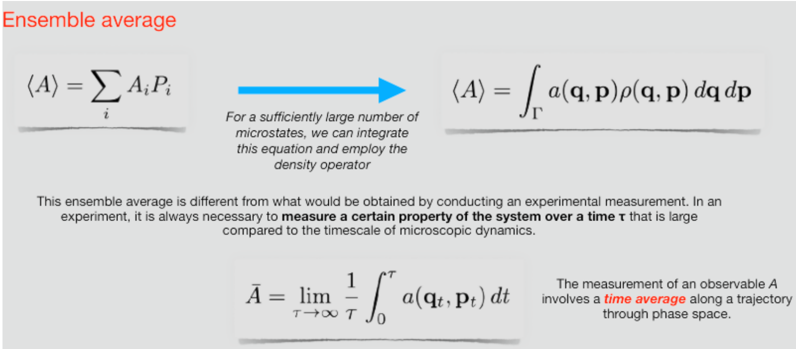
Is it always possible to find a dynamic function corresponding to a given macroscopic observable through the averaging process within a given ensemble? No, this is **not always** the case. For **mechanical** observables, like energy, pressure, and linear momentum, it's feasible to obtain them as averages of dynamic functions. However, the situation is different for **thermal** observables, such as entropy, temperature, Helmholtz or Gibbs free energy, and chemical potential. In these cases, there is no straightforward dynamic function at the microscopic level that corresponds to the macroscopic observable, as they lack microscopic counterparts. Entropy, for instance, can be expressed as:

$$S = -\int_\Gamma \ln[\rho(\mathbf{q},\mathbf{p})]\rho(\mathbf{q},\mathbf{p})\,d\mathbf{q}\,d\mathbf{p} \tag{1.11}$$

In this equation, the natural logarithm of the density function, $\ln[\rho(\mathbf{q},\mathbf{p})]$, is averaged by multiplying it with the density function itself and integrating over all possible particle positions and momenta $(\mathbf{q},\mathbf{p})$. This shows that entropy involves the averaging of the logarithm of the density function and, as a result, it doesn't have a direct microscopic equivalent. Instead, entropy is a global property of the system and is considered a thermal variable.

Temperature, another example of a thermal variable, is also related to the overall state of the system. In fact, it's only well-defined in thermodynamic equilibrium. Temperature is a property that pertains to the entire system as a whole and is not associated with any specific microscopic property. (Remember how Thermodynamics emphasises that temperature doesn't exist for individual atoms and molecules!)

1.2.3. Ensemble Average and Time Average. Ergodicity



Ensemble averaging is a statistical concept used in physics and engineering to describe the average behaviour of a system by considering all possible states or configurations within a certain macrostate. It's like taking a snapshot of a system in time and observing the different microstates it can exist in. The ensemble average of a physical quantity, denoted as $\langle A \rangle$, is calculated by summing the values of that quantity over all microstates, each weighted by the probability of the system being in that microstate. Mathematically, it is expressed as:

$$\langle A \rangle = \sum_i A_i P_i \tag{1.12}$$

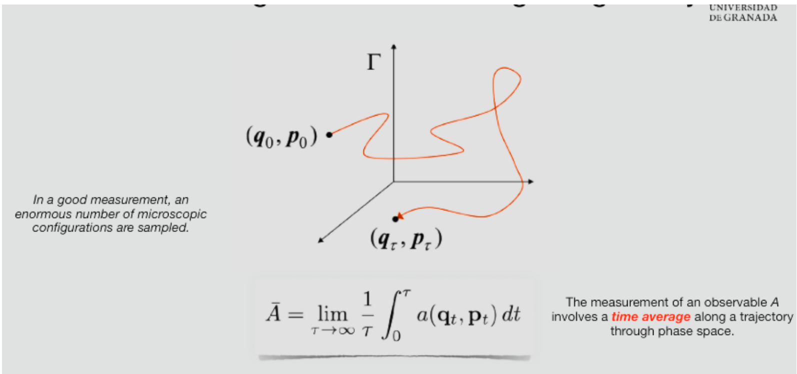
where P_i is the probability of observing a given microstate i. For a sufficiently large number of microstates, we can integrate the above equation and employ the density operator:

$$\langle A \rangle = \int_\Gamma a(\mathbf{q},\mathbf{p})\rho(\mathbf{q},\mathbf{p})\, d\mathbf{q}\, d\mathbf{p} \tag{1.13}$$

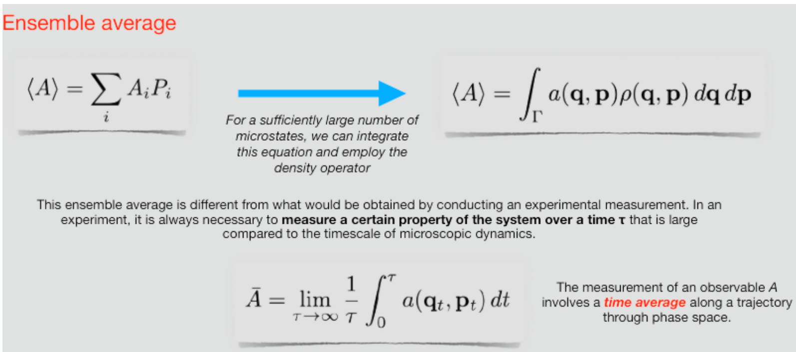
Ensemble averaging is different from the one obtained through experimental measurements. In an experiment, it's always necessary to **measure a certain property of the system over a time** τ that is significantly longer than the timescale of the microscopic dynamics. The equation below describes averaging over time by observing the system's properties over a long time interval τ.

$$\bar{A} = \mathop{\text{lím}}_{\tau\to\infty} \frac{1}{\tau}\int_0^\tau a(\mathbf{q}_t,\mathbf{p}_t)\, dt \tag{1.14}$$

Therefore, time averaging focuses on the behaviour of a **single system over time**. It involves measuring the value of a physical quantity repeatedly at different points in time and calculating the average over these measurements. In other words, measuring an observable A involves a **time average** taken along a trajectory in the phase space.



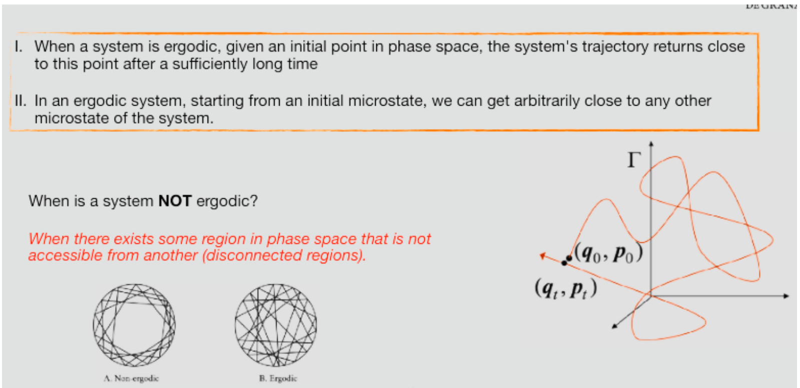
In a good measurement, an *enormous* number of microscopic configurations are traversed. For example, if you want to calculate the **time-averaged** velocity of a particle, you measure its velocity at different times and then calculate the average over those measurements. Above figure illustrates this idea. In summary, ensemble averaging considers all possible states in a macrostate and weighs them by their probabilities, while time averaging focuses on a single system's behaviour over time.



The first postulate leads us to the **ergodic hypothesis**, which states that the time average (measured) equals the ensemble average:

$$\bar{A} = \langle A \rangle = \int_\Gamma a(\mathbf{q}, \mathbf{p}) \rho(\mathbf{q}, \mathbf{p})\, d\mathbf{q}\, d\mathbf{p} \tag{1.15}$$

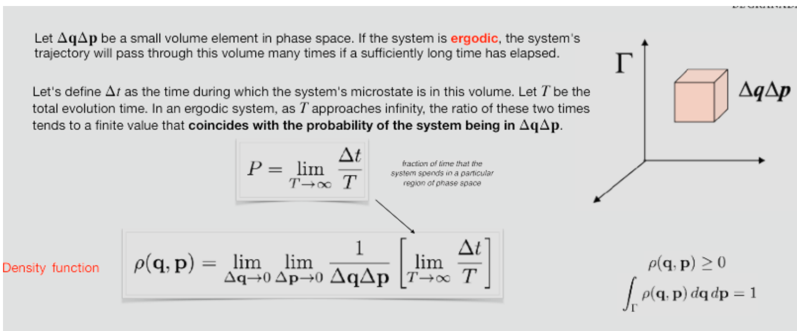
In simpler terms, the long-term time average of an observable A for a system (**measured over a sufficiently long time**) is equal to the ensemble average of that observable calculated over all possible microstates, each weighted by their probability of occurrence. In other words, the ergodicity of a system signifies that the statistical behaviour of the system can be adequately represented by time averages over a sufficiently long period. This is essential for linking the microscopic properties (captured in ensemble averages) to macroscopic observations (represented by time averages). The notion of ergodicity simplifies the analysis of complex systems since it allows us to replace the temporal evolution of a physical quantity with an ensemble average over all possible microstates. It relieves us from the need to track the real-time evolution of the physical quantity $a(\mathbf{q}_t, \mathbf{p}_t)$. This significantly eases the theoretical and practical aspects of statistical physics, especially when dealing with a vast number of interacting particles. A system that satisfies the ergodic hypothesis is referred to as an **ergodic system**. For the field of statistical physics to be applicable, it's essential that the system is ergodic.



However, it's important to remember that not all systems are ergodic, and this assumption's validity depends on the specific system being studied. When is a system non-ergodic? It's non-ergodic when there are regions in the phase space that are not accessible from others (disconnected regions).

1. In an ergodic system, given an initial point in phase space, the system's trajectory returns near this point after a sufficiently long time (see Fig. above).
2. In an ergodic system, starting from an initial microstate, one can approach any other microstate in the system as closely as desired.

1.3. Density Function. Liouville's Theorem



Consider a small portion of phase space, denoted by $\Delta\mathbf{q}\Delta\mathbf{p}$. If the system is ergodic, over a sufficiently long time, the system's trajectory will pass through this portion many times. Let's define Δt as the time during which the system's microstate is in this portion, while T is the total time of evolution. In an ergodic system, as

T tends to infinity, the ratio between these times approaches a finite value that coincides with the probability of the system being in $\Delta\mathbf{q}\Delta\mathbf{p}$:

$$P = \lim_{T\to\infty} \frac{\Delta t}{T} \tag{3.12}$$

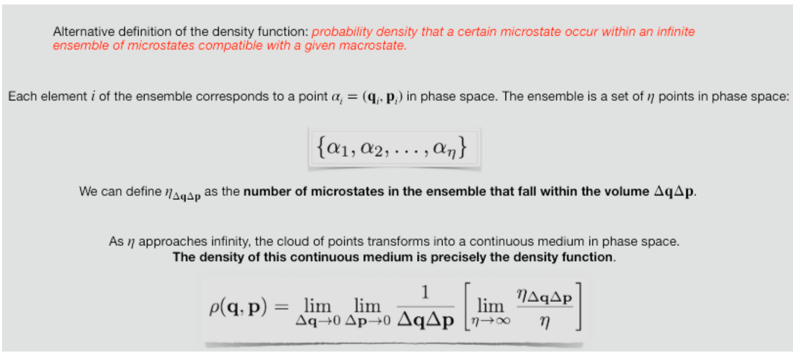
The density function can then be written as follows

$$\rho(\mathbf{q}, \mathbf{p}) = \lim_{\Delta\mathbf{q}\to 0} \lim_{\Delta\mathbf{p}\to 0} \frac{1}{\Delta\mathbf{q}\Delta\mathbf{p}} \left[\lim_{T\to\infty} \frac{\Delta t}{T} \right] \tag{1.16}$$

where $\rho(\mathbf{q}, \mathbf{p}) \geq 0$ and satisfies normalisation:

$$\int_\Gamma \rho(\mathbf{q}, \mathbf{p})\, d\mathbf{q}\, d\mathbf{p} = 1. \tag{3.14}$$

1.3.1. Alternative Definition of the Density Function



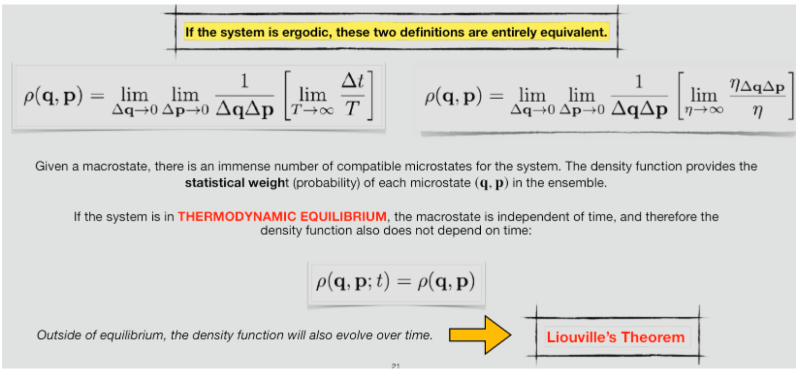
The density function can be defined as the probability density of a specific microstate occurring within an infinite ensemble of microstates compatible with a given macrostate. Each element i of the ensemble corresponds to a point $\alpha_i = (\mathbf{q}_i, \mathbf{p}_i)$ in phase space. The ensemble is a set of η points in phase space:

$$\{\alpha_1, \alpha_2, \ldots, \alpha_\eta\} \tag{1.17}$$

We can define $\eta_{\Delta\mathbf{q}\Delta\mathbf{p}}$ as the **number of microstates** in the ensemble that fall within the volume $\Delta\mathbf{q}\Delta\mathbf{p}$. As η approaches infinity, the cloud of points transforms into a continuous medium in phase space. The density of this continuous medium is precisely the density function:

$$\rho(\mathbf{q},\mathbf{p}) = \lim_{\Delta\mathbf{q}\to 0}\lim_{\Delta\mathbf{p}\to 0}\frac{1}{\Delta\mathbf{q}\Delta\mathbf{p}}\left[\lim_{\eta\to\infty}\frac{\eta_{\Delta\mathbf{q}\Delta\mathbf{p}}}{\eta}\right] \tag{1.18}$$

This density function characterises the distribution of microstates within the ensemble in the macrostate, allowing us to analyse the system's behaviour statistically.



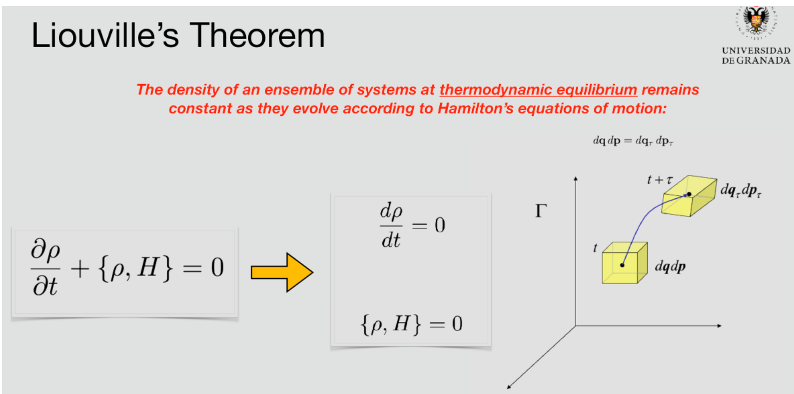
If the system is ergodic, both definitions are entirely equivalent.

Given a macrostate, there is an immense number of compatible microstates for the system. The density function provides the statistical weight (probability) of each microstate $(\mathbf{q},\mathbf{p})$ in the ensemble. If our system is in **thermodynamic equilibrium**, the macrostate is independent of time, and therefore, the density function does not depend on time either:

$$\rho(\mathbf{q},\mathbf{p};t) = \rho(\mathbf{q},\mathbf{p}) \tag{1.19}$$

Outside of equilibrium, the density function will also evolve in time. What is the time evolution of the density function? This is described by **Liouville's Theorem**.

1.3.2. Liouville's Theorem



In statistical physics, understanding the time evolution of the density function $\rho(\mathbf{q}, \mathbf{p}; t)$ is crucial. This function represents the probability density of finding the system in a specific region of phase space at time t. **Liouville's theorem** provides crucial insights into this evolution. It states that in phase space, the density of an ensemble of systems evolving under Hamilton's equations remains constant along phase-space trajectories:

$$\frac{d\rho}{dt} = \frac{\partial \rho}{\partial t} + \{\rho, H\} = 0 \tag{1.20}$$

where $\{\cdot, \cdot\}$ indicates Poisson brackets. In other words, the phase-space density is **conserved** along trajectories of the system. We notice that $\rho(\mathbf{q}, \mathbf{p}; t)$ is defined in Γ with the structure of the dynamic functions. However, while it evolves dynamically, it is not a physical observable in the usual sense, as it does not correspond to a directly measurable quantity. It should be noticed that the **total derivative** of $\rho(\mathbf{q}, \mathbf{p}; t)$ is zero, while the **partial time derivative** is zero only if the system is at the thermodynamic equilibrium (no change over time). In non-equilibrium systems, $\rho(\mathbf{q}, \mathbf{p}; t)$ evolves explicitly with time, but this evolution is exactly balanced by the system's motion in phase space, ensuring that the total derivative remains zero:

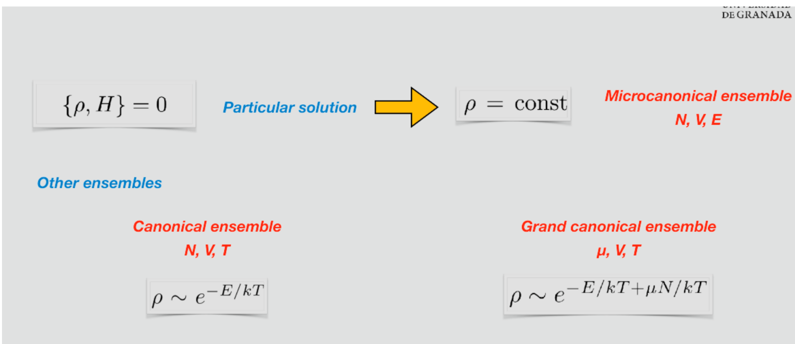
$$\frac{\partial \rho}{\partial t} + \sum_i \left(\frac{\partial \rho}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial \rho}{\partial p_i} \frac{\partial H}{\partial q_i} \right) = 0 \tag{1.21}$$

This occurs in systems such as a gas expanding into a vacuum, a system relaxing after an external perturbation, or a system driven by a time-dependent external force. By contrast, if the system is at thermodynamic equilibrium, then all the averages are time-independent and

$$\frac{\partial \rho}{\partial t} = 0 \tag{1.22}$$

which leads to

$$\{\rho, H\} = 0 \tag{1.23}$$



A particular solution of $\{\rho, H\} = 0$ is $\rho = \text{const}$. This implies that all the available states in the phase space are equally probable for the systems in that particular ensemble. Such an ensemble, where number of particles, volume and energy are constant, is referred to as **microcanonical ensemble** (it will be discussed in the following chapter). In other ensembles, ρ is stationary, but not necessarily constant. For instance, in the **canonical ensemble**, where temperature, rather than energy, is constant, $\rho \sim e^{-E/kT}$ and, in the **macrocanonical ensemble**, where the number of particles fluctuates and the chemical potential is constant along with volume and temperature, $\rho \sim e^{-E/kT+\mu N/kT}$.

Note that, in the most general case, the above result does not necessarily mean that the *shape* of a given region of phase space remains unchanged, but only that the volume this portion occupies is constant over time. In simpler terms, Liouville's theorem tells us that if we know the initial distribution of microstates in phase space (described by the density function), this distribution will remain constant as the system evolves. This theorem is particularly useful for systems obeying Hamiltonian dynamics, such as ideal gases, and plays a central role in classical statistical physics. For a detailed proof of Liouville's theorem, the interested reader is referred to Appendix E.

Application example

A free particle of mass m is constrained to move in one dimension, with a Hamiltonian given by $H = p^2/2m$. Let's verify that the area of the rectangular enclosure shown in Fig. 6.6 remains invariant during the natural evolution of the system.

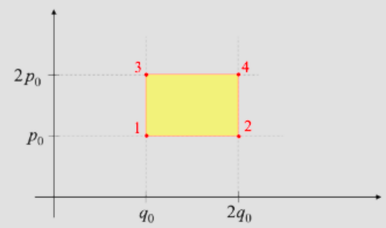


Figure 3.6: Region of the phase space where a free particle of mass m is constrained to move.

A free particle of mass m is constrained to move in one dimension, with a Hamiltonian given by $H = p^2/2m$. Let's verify that the area of the rectangular enclosure shown in Fig. above remains invariant during the natural evolution of the system.

In order to demonstrate that the area of the rectangular enclosure remains invariant during the natural evolution of the system, let's begin by defining the rectangular enclosure in the phase space and then make use of Liouville's theorem. The phase space, in this case, is two-dimensional, represented by the coordinates q and momenta p. At time $t = 0$, the rectangular enclosure can be defined in terms of its four corners:

$$(q_0, p_0), (2q_0, p_0), (q_0, 2p_0) \text{ and } (2q_0, 2p_0),$$

where $q_0, 2q_0, p_0$, and $2p_0$ are given values. The total area is then $A(t = 0) =$

$A_0 = q_0 p_0$. Let's write the equations of motion:

$$\dot{p} = -\frac{\partial H}{\partial q} = -\frac{\partial (p^2/2m)}{\partial q} = 0 \longrightarrow p = p_0 = \text{const.} \tag{1.24}$$

$$\dot{q} = \frac{\partial H}{\partial p} = \frac{\partial (p^2/2m)}{\partial p} = \frac{p}{m} \tag{1.25}$$

Integrating with respect to time, we obtain:

$$\int_0^t dq = \int_0^t \frac{p}{m} dt \longrightarrow q = q_0 + \frac{p}{m} t = q_0 + \frac{p_0}{m} t \tag{1.26}$$

The new coordinates are obtained with a linear transformation of the old coordinates. This implies that the straight lines of the rectangle are still straight. The locations of the four points at time t are:

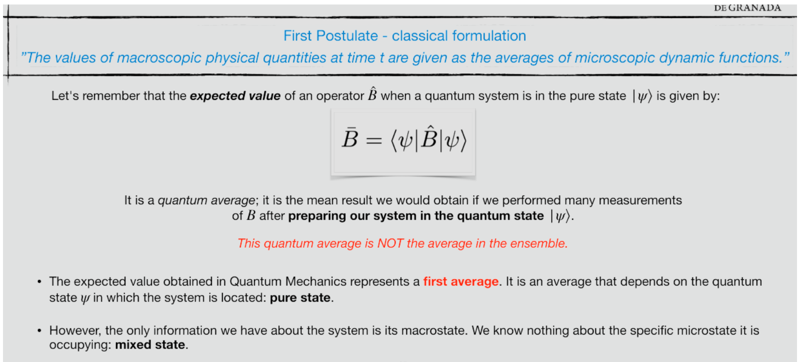
1. $a_1 = (q_0, p_0) \longrightarrow a_1' = (q_0 + \frac{p_0}{m} t, p_0)$
2. $a_2 = (2q_0, p_0) \longrightarrow a_2' = (2q_0 + \frac{p_0}{m} t, p_0)$
3. $a_3 = (q_0, 2p_0) \longrightarrow a_3' = (q_0 + \frac{2p_0}{m} t, 2p_0)$
4. $a_4 = (2q_0, 2p_0) \longrightarrow a_4' = (2q_0 + \frac{2p_0}{m} t, 2p_0)$

The area at time t can be computed as the cross product of $\vec{u} \equiv \overrightarrow{a_2' a_1'}$ and $\vec{v} \equiv \overrightarrow{a_3' a_1'}$. That is $A(t) = \vec{u} \times \vec{v} = q_0 p_0$. This result is a consequence of Liouville's theorem and demonstrates that the area of the rectangular enclosure, whose shape changes to rhombus, remains invariant during the natural evolution of the system. A quick test to corroborate the invariance of the volume (area in 2D case) of the phase space over time is showing that the Jacobian associated to the transformation $\mathcal{T} : (q_0, p_0) \to (q, p)$ is equal to 1. The Jacobian of the transformation $\mathcal{T}$ reads:

$$J = \frac{\partial (q_t, p_t)}{\partial (q_0, p_0)} = \begin{vmatrix} \frac{\partial q_t}{\partial q_0} & \frac{\partial q_t}{\partial p_0} \\ \frac{\partial p_t}{\partial q_0} & \frac{\partial p_t}{\partial p_0} \end{vmatrix} = \begin{vmatrix} 1 & \frac{t}{m} \\ 0 & 1 \end{vmatrix} = 1 \tag{1.27}$$

We conclude that the Jacobian is indeed equal to 1 and consequently the transformation $\mathcal{T}$ satisfies Liouville's theorem. However, it should be noticed that calculating the Jacobian says nothing on whether the shape of the phase space changes or not.

1.4. Quantum Formulation of the First Postulate



As we mentioned in the previous section, the first postulate of classical statistical mechanics states that the state of a system is described by a probability distribution $\rho(\mathbf{q}, \mathbf{p})$ over its microstates, where $\mathbf{q}$ and $\mathbf{p}$ are the generalized coordinates and momenta of the system. The macroscopic properties of the system, such as the expected value of an observable $B(\mathbf{q}, \mathbf{p})$, are calculated as ensemble averages, which are integrals over phase space:

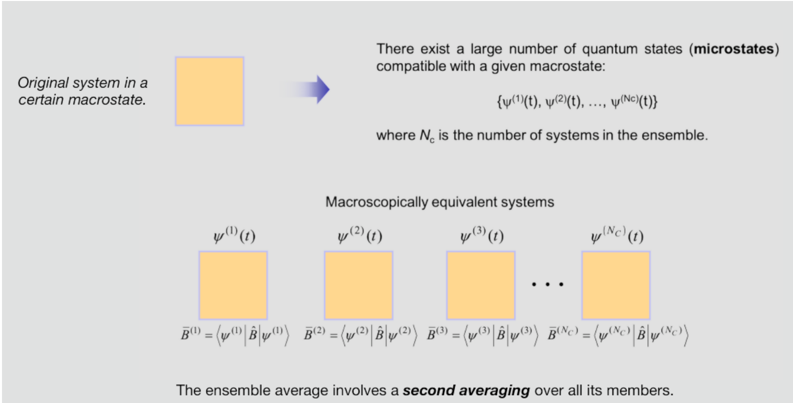
$$\langle B \rangle = \int_\Gamma B(\mathbf{q}, \mathbf{p})\rho(\mathbf{q}, \mathbf{p})\, d\mathbf{q}d\mathbf{p} \tag{1.28}$$

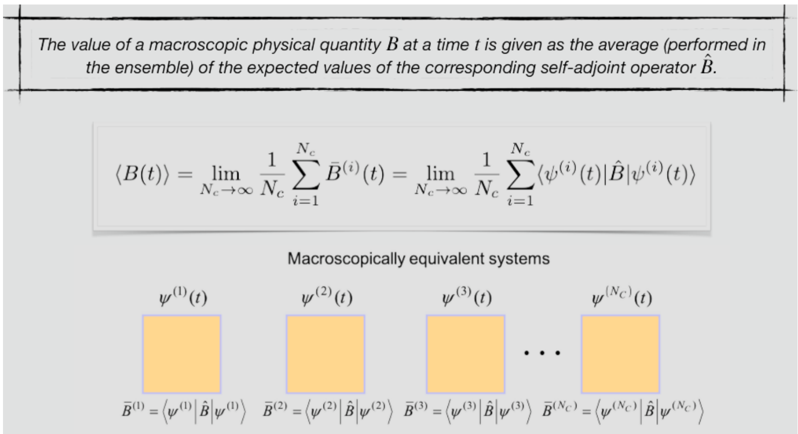
Now, consider a quantum system prepared in the pure state $|\psi\rangle$. The expected value of an operator $\hat{B}$ in this state is given by

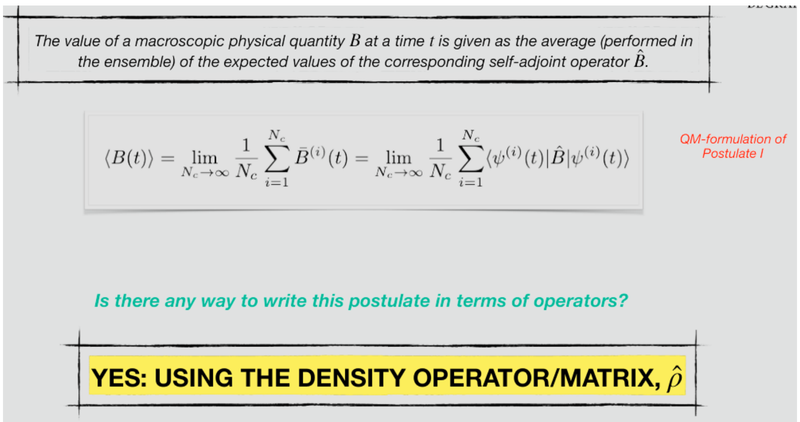
$$\bar{B} = \langle\psi|\hat{B}|\psi\rangle = \int \Psi^*(q_1, \ldots, q_s)(\hat{B}\Psi(q_1, \ldots, q_s))\, dq_1 \ldots dq_s \tag{1.29}$$

where $\bar{B}$ is the quantum expectation value of the observable B, $|\psi\rangle$ represents the quantum state of the system (a pure state), $\hat{B}$ is the operator associated with the observable B, $\Psi(q_1, \ldots, q_s)$ is the wavefunction of the system in the chosen representation and $\Psi^*(q_1, \ldots, q_s)$ is the complex conjugate of the wavefunction. This expectation value represents the mean outcome obtained from repeated measurements of B on identically prepared systems in the state $|\psi\rangle$. It is essential to distinguish this **quantum average** from an ensemble average, which involves a second average.

In Quantum Mechanics, the expectation value is a first-level average, determined solely by the quantum state ψ (a pure state). However, in practical scenarios, we often lack complete knowledge of the system's exact microstate. Instead, we only have information about its macrostate, which may correspond to a statistical mixture of pure states, that is a mixed state.







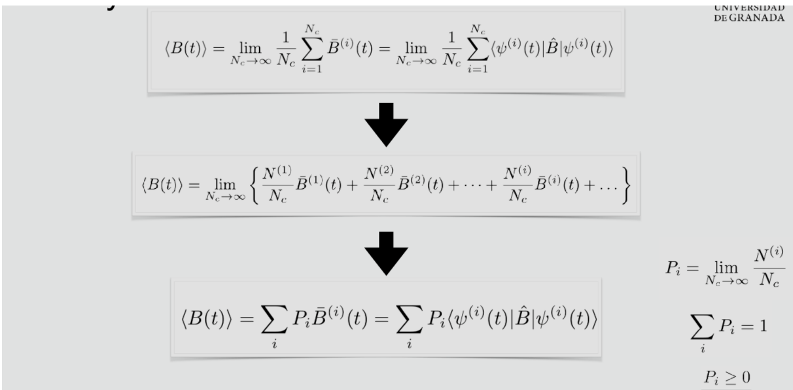
In the case of an ensemble consisting of a large number of systems, the ensemble average of an operator $\hat{B}$ at time t is given by:

$$\langle B(t) \rangle = \lim_{N_c \to \infty} \frac{1}{N_c} \sum_{i=1}^{N_c} \bar{B}^{(i)}(t) = \lim_{N_c \to \infty} \frac{1}{N_c} \sum_{i=1}^{N_c} \langle \psi^{(i)}(t) | \hat{B} | \psi^{(i)}(t) \rangle \tag{1.30}$$

This equation states that, in an ensemble of N_c quantum systems, the macroscopic physical quantity B at time t is obtained by averaging the expectation values of the corresponding operator B over all individual systems. In the limit as $N_c \to \infty$, this ensemble average provides a **statistical description** of the system's macroscopic behaviour. A more compact and general way to express this ensemble average is through the **density operator** (or **density matrix**) $\hat{\rho}$, which describes the statistical mixture of quantum states within an ensemble. We will see how in the following section. Thus, Eq. (1.30) emphasises that, when dealing with a large number of systems, the expectation value of an operator (representing a macroscopic physical quantity) can be equivalently obtained as an ensemble average.

In summary, the first postulate in quantum mechanics involves (i) a quantum average that describes the behaviour of individual quantum systems (specific pure states), and (ii) a statistical average that describes the overall system when it is in a mixed state or an ensemble of quantum systems.

1.5. Density Matrix. Von Neumann's Equation



The introduction of the density matrix provides a powerful framework for describing quantum systems statistically, bridging the gap between quantum mechanics and thermodynamic ensembles. When we describe a system using a density matrix, it means that the system is not necessarily in a single, well-defined quantum state (a pure state). Instead, it could be in one of several pure states $|\psi^{(i)}(t)\rangle$, each with a probability P_i. Basically, the density matrix encodes both the quantum mechanical superposition (within each $|\psi^{(i)}(t)\rangle$) and the classical uncertainty (the probability P_i) over which state the system is actually in. This is what we are discussing in this section.

In the average of Eq. 3.28, certain systems might be in the same quantum state $|\psi^{(i)}\rangle$. For instance, $N^{(1)}$ systems in $|\psi^{(1)}\rangle$, $N^{(2)}$ systems in $|\psi^{(2)}\rangle$, etc. It follows that

$$\langle B(t)\rangle = \text{lím}_{N_c\to\infty} \left\{ \frac{N^{(1)}}{N_c}\bar{B}^{(1)}(t) + \frac{N^{(2)}}{N_c}\bar{B}^{(2)}(t) + \cdots + \frac{N^{(i)}}{N_c}\bar{B}^{(i)}(t) + \dots \right\} \tag{1.31}$$

Let's define the probability to find the quantum state $|\psi^{(i)}\rangle$ in the ensemble as

$$P_i = \lim_{N_c \to \infty} \frac{N^{(i)}}{N_c} \tag{1.32}$$

Then Eq. (1.31) can be rewritten as

$$\langle B(t) \rangle = \sum_i P_i \bar{B}^{(i)}(t) = \sum_i P_i \langle \psi^{(i)}(t) | \hat{B} | \psi^{(i)}(t) \rangle \tag{1.33}$$

with

$$\sum_i P_i = 1 \tag{1.34}$$

and

$$P_i \geq 0 \tag{1.35}$$

To obtain the matrix expression of the operator, it is necessary to write all the functions $\psi^{(i)}(t)$ in the orthonormal basis $\{|\phi_n\rangle\}$ of the Hilbert space:

$$|\psi^{(i)}(t)\rangle = \sum_n c_n^{(i)}(t) |\phi_n\rangle$$

$$\langle \psi^{(i)}(t)| = \sum_n c_n^{(i),*}(t) \langle \phi_n|$$

Some steps...

$$\bar{B}^{(i)}(t) = \sum_{n,m} c_n^{(i),*}(t)\, c_m^{(i)}(t) \underbrace{\langle \phi_n | \hat{B} | \phi_m \rangle}_{B_{nm}}$$

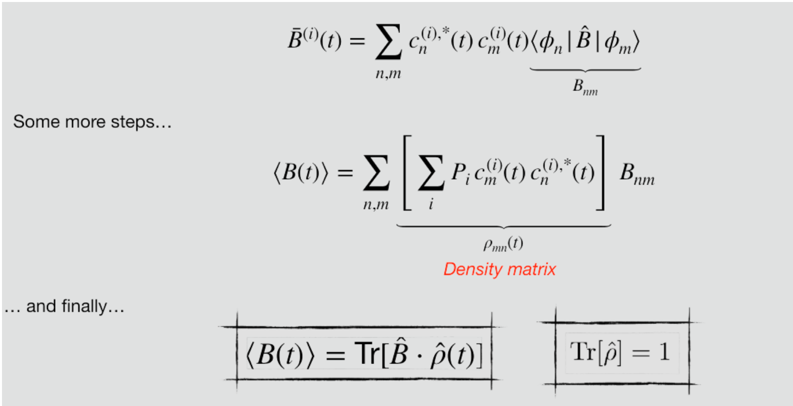
The state vectors $|\psi^{(i)}(t)\rangle$ are normalized to unity, although they may not necessarily be orthogonal to one another. To obtain the matrix representation of the operator, it is necessary to express each state vector $|\psi^{(i)}(t)\rangle$ in an orthonormal basis $\{|\phi_n\rangle\}$ of the Hilbert space:

$$|\psi^{(i)}(t)\rangle = \sum_n c_n^{(i)} |\phi_n\rangle \tag{1.36}$$

$$\langle \psi^{(i)}(t)| = \sum_n c_n^{(i)*} \langle \phi_n| \tag{1.37}$$

where $c_n^{(i)} = c_n^{(i)}(t)$ and $c_n^{(i)*} = c_n^{(i)*}(t)$ are complex numbers. It follows that [2]

$$
\begin{aligned}
\bar{B}^{(i)}(t) &= \langle \psi^{(i)}(t)|\hat{B}|\psi^{(i)}(t)\rangle = \\
&= \left\{ \sum_n c_n^{(i)*} \langle \phi_n| \right\} \hat{B} \left\{ \sum_m c_m^{(i)} |\phi_m\rangle \right\} = \\
&= \sum_{n,m} c_n^{(i)*} c_m^{(i)} \underbrace{\langle \phi_n|\hat{B}|\phi_m\rangle}_{B_{nm}}
\end{aligned}
\tag{1.38}
$$



In particular, B_{nm} is the matrix element of $\hat{B}$ in the $\{|\phi_n\rangle\}$ basis, which gives the inner product between the state $|\phi_n\rangle$ and the result of applying the operator $\hat{B}$ to the state $|\phi_m\rangle$. It follows that the ensemble average reads

$$
\langle B(t) \rangle = \sum_i P_i \bar{B}^{(i)}(t) = \sum_i P_i \sum_{n,m} c_n^{(i)*} c_m^{(i)} B_{nm}
\tag{1.39}
$$

where the first sum on the right-hand side is over different elements of the ensemble, whereas the second sum takes place in the orthonormal basis $\{|\phi_n\rangle\}$. Let's rearrange

$$
\langle B(t) \rangle = \sum_{n,m} \left[\underbrace{\sum_i P_i c_m^{(i)} c_n^{(i)*}}_{\rho_{mn}(t)} \right] B_{nm}
\tag{1.40}
$$

where $\rho_{mn}(t)$ is the **density matrix**. Now, the density matrix $\rho_{mn}(t)$ is not generally diagonal. However, there always exists a basis in which it is diagonal. We can assume that the orthonormal basis $\{|\phi_n\rangle\}$ we have chosen is one such basis (see also observations below). It follows that

$$\langle B(t)\rangle = \sum_{n,m}\rho_{mn}(t)B_{nm} = \sum_n \underbrace{\left[\sum_m B_{nm}\rho_{mn}(t)\right]}_{(\hat{B}\cdot\hat{\rho}(t))_{nn}} \tag{1.41}$$

The last equation can be further simplified

$$\langle B(t)\rangle = \sum_n (\hat{B}\cdot\hat{\rho}(t))_{nn} = \mathrm{Tr}[\hat{B}\cdot\hat{\rho}(t)] \tag{1.42}$$

- It is self-adjoint $\longrightarrow$ $\rho^*_{nm} = \rho_{mn}$
- The diagonal elements are probabilities of finding a system in the ensemble in the state $\phi^{(n)}$ $\longrightarrow$ $\rho_{nn} = \sum_i P_i\,|c^{(i)}_{nn}|^2 = P_n$
- Always, there exists an orthonormal basis that makes the density operator diagonal: $\longrightarrow$ $\rho_{mn} = P_n\delta_{mn}$
- The diagonal elements are positive and satisfy: $\longrightarrow$ $\sum_n \rho_{nn} = 1 \;\Rightarrow\; 0 \le \rho_{nn} \le 1$

Some observations:

- The density operator is self-adjoint: $\rho^*_{nm} = \rho_{mn}$. Proof:

$$\rho^*_{nm} = \left(\sum_i P_i c^{(i)}_n(t)c^{(i)*}_m(t)\right)^* = \sum_i P_i c^{(i)*}_n(t)c^{(i)}_m(t) = \rho_{mn}(t) \tag{1.43}$$

- The elements on the diagonal of the density matrix $\rho_{mn}(t)$ represent the probability P_n of finding a system in the ensemble in the state $|\psi^{(n)}(t)\rangle$:

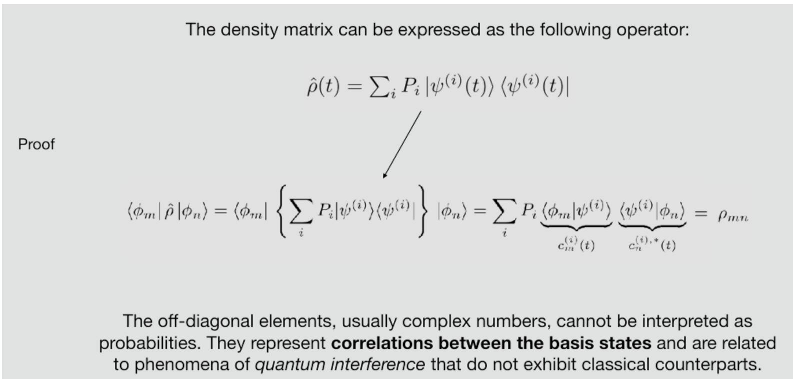
$$\rho_{nn}(t) = \sum_i P_i|c^{(i)}_n(t)|^2 = P_n \tag{1.44}$$

with $0 \le \rho_{nn}(t) \le 1$. The quantity P_i represents the probability that the system is in the pure quantum state $|\psi^{(i)}(t)\rangle$ within a statistical ensemble. This is a classical (statistical) probability, indicating that the system is in one of the states

$|\psi^{(i)}(t)\rangle$ with probability P_i, but without specifying which particular state. On the other hand, $|c_n^{(i)}(t)|^2$ represents the probability that a system, already in the pure state$|\psi^{(i)}(t)\rangle$, is found in the basis state $|\phi_n\rangle$ upon measurement. This is a quantum probability, arising from the superposition principle and the probabilistic nature of quantum mechanics[4]. It follows that $\sum_n |c_n^{(i)}(t)|^2 = 1$, and thus,

$$\sum_n \rho_{nn}(t) = \sum_n \sum_i P_i |c_n^{(i)}(t)|^2 = \sum_i P_i \sum_n |c_n^{(i)}(t)|^2 = \sum_i P_i = 1 \longrightarrow \mathrm{Tr}[\hat{\rho}(t)] = 1 \tag{1.45}$$

Thus, while P_i describes the statistical uncertainty regarding which pure state the system occupies, $|c_n^{(i)}(t)|^2$ describes the intrinsic quantum uncertainty in the measurement of a system already in a specific pure state. Combining these two probabilities, the density matrix formalism provides a complete description of both classical and quantum uncertainties in statistical quantum mechanics.



- In general, the density matrix is not diagonal. The off-diagonal elements, usually complex numbers, cannot be interpreted as probabilities. They represent correlations between the basis states and are related to phenomena of **quantum interference** that do not exhibit classical counterparts. However, there always exists an orthonormal basis that makes the density operator diagonal: $\rho_{mn}(t) = P_n \delta_{mn}$.

- The density matrix can be expressed as an operator, that is $\hat{\rho}(t) = \sum_i P_i |\psi^{(i)}(t)\rangle\langle\psi^{(i)}(t)|$. This definition accounts for both the quantum state and the statistical probabilities. Let's prove this result by projecting onto an orthonormal basis $\{|\phi_n\rangle\}$ and finding the matrix elements of $\hat{\rho}$ in this basis:

$$\langle\phi_m|\hat{\rho}(t)|\phi_n\rangle = \langle\phi_m| \left\{ \sum_i P_i |\psi^{(i)}(t)\rangle\langle\psi^{(i)}(t)| \right\} |\phi_n\rangle = \sum_i P_i \underbrace{\langle\phi_m|\psi^{(i)}(t)\rangle}_{c_m^{(i)}} \underbrace{\langle\psi^{(i)}(t)|\phi_n\rangle}_{c_n^{(i)*}} = \rho_{mn}(t) \tag{1.46}$$

We then recover the definition of the density matrix $\rho_{mn}(t)$ as introduced in Eq. 3.38.

- If $\{|\phi_n\rangle\}$ is a basis that diagonalizes the density operator, then the expression for the average is simpler:

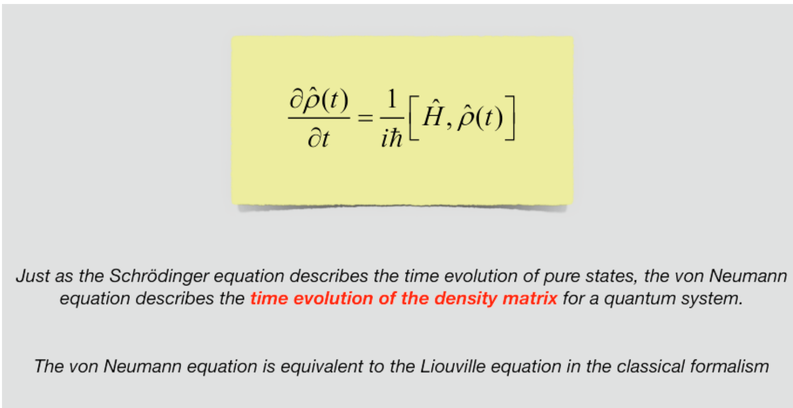
$$\langle B(t)\rangle = \mathrm{Tr}[\hat{\rho}(t)\cdot\hat{B}] = \sum_{nm}\rho_{mn}(t)B_{nm} = \sum_{nm}P_n\delta_{nm}B_{nm} = \sum_{n}P_nB_{nn} \quad (1.47)$$

It follows that

$$\langle B(t)\rangle = \sum_{n}P_n(t)\underbrace{\langle\phi_n|\hat{B}|\phi_n\rangle}_{B_{nn}} \quad (1.48)$$

where B_{nn} is the expected value of $\hat{B}$ in the basis $\{|\phi_n\rangle\}$.

von Neumann Equation



If $|\psi^{(i)}(t)\rangle$ is a pure state corresponding to the state i of the ensemble, then it must follow the Schrödinger equation:

$$i\hbar\frac{\partial}{\partial t}|\psi^{(i)}(t)\rangle = \hat{H}|\psi^{(i)}(t)\rangle \quad (1.49)$$

Equivalently

$$i\hbar\frac{\partial}{\partial t}\langle\psi^{(i)}(t)| = -\langle\psi^{(i)}(t)|\hat{H} \tag{1.50}$$

$$i\hbar\frac{\partial\hat{\rho}(t)}{\partial t} = i\hbar\frac{\partial}{\partial t}\left\{\sum_i P_i|\psi^{(i)}(t)\rangle\langle\psi^{(i)}(t)|\right\} = i\hbar\sum_i P_i\left\{\frac{\partial|\psi^{(i)}(t)\rangle}{\partial t}\langle\psi^{(i)}(t)| + |\psi^{(i)}(t)\rangle\frac{\partial\langle\psi^{(i)}(t)|}{\partial t}\right\} \tag{1.51}$$

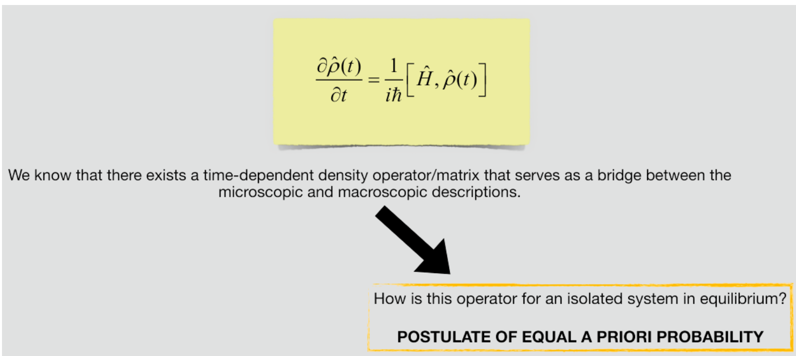
which gives

$$i\hbar\frac{\partial\hat{\rho}(t)}{\partial t} = \sum_i P_i\left\{\hat{H}|\psi^{(i)}(t)\rangle\langle\psi^{(i)}(t)| - |\psi^{(i)}(t)\rangle\langle\psi^{(i)}(t)|\hat{H}\right\} = \hat{H}\hat{\rho} - \hat{\rho}\hat{H} = [\hat{H},\hat{\rho}] \tag{1.52}$$

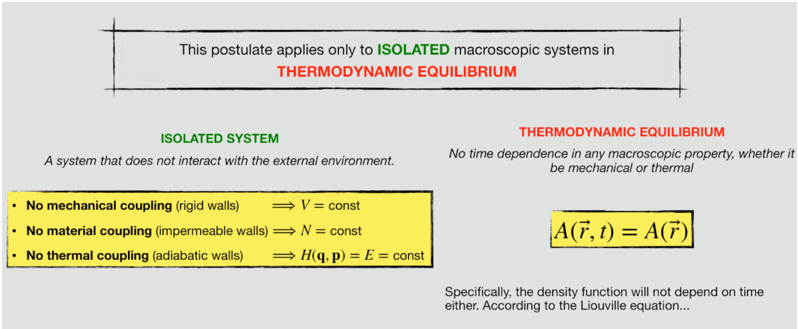
In its final form:

$$\frac{\partial\hat{\rho}(t)}{\partial t} = \frac{1}{i\hbar}[\hat{H},\hat{\rho}] \tag{1.53}$$

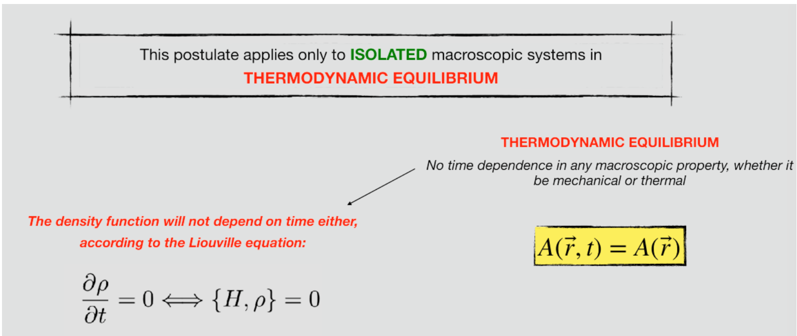
where $[\hat{H},\hat{\rho}] = \hat{H}\hat{\rho} - \hat{\rho}\hat{H}$ is the **commutator**. Eq. (1.53) is known as the **von Neumann equation**. Just as the Schrödinger equation describes the time evolution of pure states, the von Neumann equation, also known as the Liouville-von Neumann equation or the quantum master equation, describes the time evolution of the density matrix for a quantum system. **It is equivalent to the Liouville equation in the classical formalism**: $\partial\rho/\partial t = \{H,\rho\}$ (see Appendix E).



1.6. Postulate II: Equal Prior Probability



We now know that there exists a density operator or matrix that evolves over time, linking the microscopic and macroscopic descriptions of a physical system. This operator plays a crucial role in our understanding of the system's behaviour. Now, let's explore the specific characteristics of this operator when applied to an **isolated system in equilibrium**. This brings us to the *Postulate of Equal A Priori Probability*, shedding light on how probabilities are distributed in such systems. This postulate applies exclusively to macroscopic systems that are both **isolated** and in **thermodynamic equilibrium**.



An isolated system is one that remains entirely independent of its surroundings, with no external interactions. This means that the number of particles (or molecules, atoms, etc.) (N), volume (V) and the Hamiltonian (H) of the system, which describes its total energy, remain constant. More specifically:

- No mechanical coupling through rigid walls: V constant.
- No material exchange through impermeable walls: N constant.
- No thermal interchange through adiabatic walls: $H(\mathbf{q},\mathbf{p}) = E$ constant.

In the classical description, thermodynamic equilibrium is characterised by the absence of any time dependence in any macroscopic property, whether it be mechanical or thermal ($A(\vec{r},t) = A(\vec{r})$). Specifically, even the density function (or density matrix) remains independent of time. This property is rigorously expressed through the Liouville equation:

$$\frac{\partial \rho}{\partial t} = 0 \iff \{\rho, H\} = 0 \tag{1.54}$$